

# Sarah Connor

AI Security Engineer · Adversarial ML & Red Teaming

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## > SUMMARY

AI security engineer with six years spanning applied machine learning, security engineering, and detection engineering. Builds defenses against adversarial and prompt-injection attacks on production ML systems, and leads red-team exercises against internal LLM deployments.

## > EXPERIENCE



### Senior AI Security Engineer

Cyberdyne Systems

2023 – Present

Austin, USA

Leading red-team engagements against internal LLM-powered products before launch.

- Built an automated prompt-injection fuzzing harness that surfaced 40+ high-severity jailbreaks across 6 production LLM features prior to release.  
Custom mutation engine over a 12k-seed corpus, wired into the release-gate CI job; findings triaged weekly with the app-sec team.
- Led adversarial robustness reviews for a fraud-detection model serving 2M inference requests/day, reducing evasion success rate from 18% to under 2%.
- Designed and shipped a model-extraction detection pipeline (query-pattern anomaly scoring) now running across all customer-facing inference endpoints.

Python · PyTorch · Burp Suite · Kubernetes · OPA/Gatekeeper



### Security Engineer, Applied ML

Skynet Analytics

2021 – 2023

Remote

Promoted from IC to owning the security review process org-wide within eight months.

Owned the security review process for the ML platform team's model deployment pipeline.

- Authored the org's first threat model for LLM-integrated services, adopted as the standard review template company-wide.
- Cut mean time to patch critical model-serving CVEs from 21 days to 4 by embedding security gating directly into the MLOps CI/CD pipeline.
- Ran quarterly red-team exercises against internal chatbots, discovering data-exfiltration paths later closed via output filtering and scoped tool permissions.

Terraform · AWS · MLflow · Semgrep · Snyk



### Machine Learning Engineer

Tyrell ML

2019 – 2021

Ljubljana, Slovenia

Built and shipped machine learning models powering fraud detection and demand forecasting for a fast-growing fintech product.

- Built and shipped fraud-detection models into production, reducing false-positive rates by 25% across the company's core payments pipeline.
- Designed the team's first automated retraining pipeline, cutting model refresh time from weeks to days.

TensorFlow · scikit-learn · Docker

## > SKILLS

|          |                                                                     |
|----------|---------------------------------------------------------------------|
| Security | Threat modeling, Burp Suite, Nmap, Metasploit                       |
| ML/AI    | PyTorch, vLLM, adversarial robustness, prompt-injection defense     |
| Infra    | Kubernetes, Terraform, AWS, CI/CD pipeline hardening                |
| Practice | Incident response, vulnerability research, red-team exercise design |

## > EDUCATION

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### **M.S. Computer Security**

Stanford University

2023

Stanford, USA



### **B.S. Computer Science**

University of Ljubljana

2019

Ljubljana, Slovenia

## > PROJECTS

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### **Prompt Injection Firewall**

[github.com/sconnor/promptfirewall](https://github.com/sconnor/promptfirewall)

2023

## > CERTIFICATIONS, AWARDS & PUBLICATIONS

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### **OSCP – Offensive Security Certified Professional**

Offensive Security

2019



### **AI Village CTF – 1st Place**

DEF CON 24, Las Vegas, USA

2024



### **Detecting Prompt Injection at Scale**

[arxiv.org/abs/2404.16244](https://arxiv.org/abs/2404.16244)

2024